Baseline

A RESEARCH AID FOR MEASURING LUNG NODULES ON CT

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A click outlines a lung nodule on CT; checking, correcting and recording it takes minutes, not half an hour.

Dr. Daniel Reyes of the Thoracic Oncology Research Lab at Harbor Medical Center (a simulated lab, for a home assignment) measures lung nodules by hand for a growth study. By his own step times, a scan with a volume takes 20 to 34 minutes, so he skips the volume on most tumours. With Baseline, he clicks the nodule, checks and corrects the 3D outline, and records it in a shared Study table. For the first read, a nodule finder acts as a co-pilot: it marks places worth a look, and he still reads the whole scan.

WHAT WE MEASURED

2.3 min
per scan, against 20–34 min by hand

78.5%
less time measuring and recording (projected week)

15 of 18
within the expert's re-measure range after a median of 2 corrections (one click: 5 of 18)

15–20 s
to re-find a nodule on the follow-up, against about 10 min

18 timed runs on 9 public patients by one operator. A script placed the corrections, so read these as best cases. Details: results/.

WORKING WITH THE LAB

THE LAB TOLD US	SO WE
Daniel: finding is his slowest step, and he skips the volume on 4 of 5 tumours "because it takes so much time".	**built measuring first, so every tumour gets a volume, then the finder.**
Naomi: success is "correcting rather than doing the full tracing".	**timed whole scans, corrections included.**
Daniel: stay "within the range I see when I re-measure the same nodule".	**made that range the quality bar.**
Naomi: standard equipment is "much more important" than high-end hardware.	**run it on a PC with no graphics card (about 20 s per click).**
Daniel "always" includes the hazy ground glass.	**set the paint tool to keep it.**
Naomi keeps her spreadsheet as the main record.	**kept the CSV as the main path, with the Study table as backup.**

TRY IT IN THREE MINUTES

1. Sign in at lung-measure.base44.app (invite only).
2. Choose **Use the practice server**, then **Start the tutorial**.
3. Optional: one-command install at github.com/hillel-base/baseline.

METHODS

The outline: nnInteractive

A 3D promptable segmentation model. One click gives the nodule's outline on every slice, Add and Remove clicks correct it, and the tool measures the volume and axes from the outline. Hard nodules (hazy edges, contact with other tissue, spikes) need corrections. A click takes about 20 s on a standard PC and about 1 s on a graphics chip.

Isensee F., Rokuss M., Krämer L. et al., *nnInteractive: Redefining 3D Promptable Segmentation*, 2025, arXiv:2503.08373 · code github.com/MIC-DKFZ/nnInteractive · weights huggingface.co/MIC-DKFZ/nnInteractive (CC BY-NC-SA 4.0, research use only).

The nodule finder: a lightweight 2.5D detector

It marks places worth a look, and the reader still reads the whole scan. On 50 public screening scans it found about 7 in 10 of the nodules that at least 3 of 4 radiologists marked, with about 2 extra marks per scan, in about 22 s. It can miss nodules larger than 3 cm.

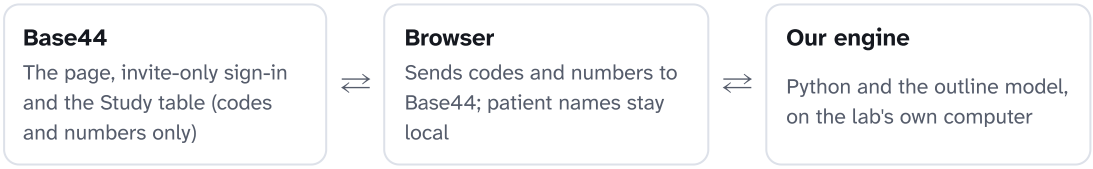
Chou L.-F. et al., *Diagnostics* 2026;16(18):3038, doi:10.3390/diagnostics16183038 · weights huggingface.co/Lien-Feng/Lightweight-2-5D-LUNA16 (CC BY 4.0), trained on LUNA16 (from LIDC-IDRI) · built on Ultralytics YOLO11 (AGPL-3.0) · source baseline/finder.

Follow-up matching: whole-chest alignment

The tool aligns the repeat scan to the first one (rigid, then affine, by mutual information; 4 to 12 s on the processor) and marks where each earlier nodule should be, for the reader to confirm. Within 1 cm: 90 of 93 nodules on screening follow-ups a year apart, and 26 of 28 on our same-day pairs.

SimpleITK image registration (simpleitk.org); tested on public NLST follow-ups and RIDER.

HOW IT IS BUILT



The scans never leave the lab's computer. For this review, two practice servers run the same engine on public scans.

HOW I WORKED

Six Claude Code agents did the work with me, each in its own session with one role. They hand work to each other through a shared board in the repository.

Strategist
plans and records decisions

Liaison
drafts my messages to the lab

Advisor
researches and runs quick tests

Builder
writes the code, by pull request

Evaluator
measures what the Builder ships

Guide
teaches me the imaging

LINKS

The tool: lung-measure.base44.app (invite only)

The source and the one-command install: github.com/hillel-base/baseline

The results, scripts and data: baseline/results

DATA

From TCIA: RIDER Lung CT (CC BY 4.0) and RIDER-LungCT-Seg (CC BY 3.0); QIBA VolCT 1B radiologist measurements (CC BY 4.0); NLST and LIDC-annot-NLST501 (CC BY 4.0). Also the LUNA25 nodule annotations (CC BY-NC 4.0).